Unit: Unit 2: Exponents Radicals **Name:** ________________
Topic: Rationalizing Denominator **Date:** ____________

Advanced (Habanero)

Q1. Rationalize the denominator: $\frac{1}{\sqrt{2}}$

(A) $\frac{\sqrt{2}}{2}$
(B) $\frac{1}{2\sqrt{2}}$
(C) $\frac{2}{\sqrt{2}}$
(D) $\frac{\sqrt{2}}{4}$
(E) $\frac{1}{2}$

Q2. Simplify: $\frac{1}{\sqrt{5}+1}$

(A) $\frac{\sqrt{5}-1}{4}$
(B) $\frac{\sqrt{5}+1}{4}$
(C) $\frac{1}{\sqrt{5}-1}$
(D) $\frac{\sqrt{5}-1}{5-1}$
(E) $\frac{\sqrt{5}+1}{5+1}$

Q3. Rationalize: $\frac{2}{\sqrt{3}-2}$

(A) $\frac{2(\sqrt{3}+2)}{3-4}$
(B) $\frac{2(\sqrt{3}+2)}{-1}$
(C) $\frac{2(\sqrt{3}-2)}{3-4}$
(D) $\frac{2(\sqrt{3}+2)}{1}$
(E) $\frac{2(\sqrt{3}+2)}{-3}$

Q4. Simplify: $\frac{\sqrt{7}}{\sqrt{7}+\sqrt{3}}$

(A) $\frac{\sqrt{7}-\sqrt{3}}{7-3}$
(B) $\frac{\sqrt{7}-\sqrt{3}}{7+3}$
(C) $\frac{\sqrt{7}+\sqrt{3}}{7+3}$
(D) $\frac{\sqrt{7}+\sqrt{3}}{7-3}$
(E) $\frac{\sqrt{7}+\sqrt{3}}{7}$

Q5. Rationalize: $\frac{3}{\sqrt{2}+\sqrt{5}}$

(A) $\frac{3(\sqrt{2}-\sqrt{5})}{2-5}$
(B) $\frac{3(\sqrt{2}-\sqrt{5})}{2+5}$
(C) $\frac{3(\sqrt{2}+\sqrt{5})}{2-5}$
(D) $\frac{3(\sqrt{2}-\sqrt{5})}{-3}$
(E) $\frac{3(\sqrt{2}+\sqrt{5})}{2+5}$

Q6. Simplify: $\frac{\sqrt{6}}{\sqrt{2}-\sqrt{3}}$

(A) $\frac{\sqrt{6}(\sqrt{2}+\sqrt{3})}{2-3}$
(B) $\frac{\sqrt{6}(\sqrt{2}-\sqrt{3})}{2+3}$
(C) $\frac{\sqrt{6}(\sqrt{2}+\sqrt{3})}{2+3}$
(D) $\frac{\sqrt{6}(\sqrt{2}-\sqrt{3})}{2-3}$
(E) $\frac{\sqrt{6}(\sqrt{2}-\sqrt{3})}{2}$

Q7. Rationalize: $\frac{1}{\sqrt{10}-\sqrt{2}}$

(A) $\frac{\sqrt{10}+\sqrt{2}}{10-2}$
(B) $\frac{\sqrt{10}-\sqrt{2}}{10+2}$
(C) $\frac{\sqrt{10}-\sqrt{2}}{10-2}$
(D) $\frac{\sqrt{10}+\sqrt{2}}{10+2}$
(E) $\frac{\sqrt{10}+\sqrt{2}}{8}$

Q8. Simplify: $\frac{2}{\sqrt{5}-2}$

(A) $\frac{2(\sqrt{5}+2)}{5-4}$
(B) $\frac{2(\sqrt{5}-2)}{5+4}$
(C) $\frac{2(\sqrt{5}+2)}{5+4}$
(D) $\frac{2(\sqrt{5}+2)}{-1}$
(E) $\frac{2(\sqrt{5}-2)}{5-4}$

Open Ended Questions (FRQ)

Q9. Rationalize the denominator and simplify: $\frac{3\cdot\sqrt{2}}{\sqrt{6}-\sqrt{3}}$

Q10. Rationalize the denominator: $\frac{\sqrt{7}-\sqrt{3}}{\sqrt{7}+\sqrt{3}}$

Chef's Tips

- Tip 1: Make sure to simplify the rationalized form.
- Tip 2: Check if the denominator can be further simplified.
- Tip 3: Use the conjugate to rationalize the denominator.